IMPORTANT NOTICE:

This round consists of two documents. The second document is available in the “Download Attachments” section along with the Round-1 Rulebook. Evaluation will be based on both documents.

Aurex Dynamics Private Limited

Internal Audit Report: Storyline & Structure Review

Confidential | Prepared for MES 2026 – Blind Spot Case Exercise

Scope & Context

This document forms part of an internal-style audit intended to evaluate the Storyline and **Structure** of Aurex Dynamics Private Limited as the company transitions from early-stage execution into institutional scale. The review is structured to resemble the kind of assessment typically commissioned by venture capital investors or large municipal partners prior to long-term engagement.

The audit does not assume malicious intent, misrepresentation, or negligence. Instead, it captures how fast-growing, founder-led engineering companies often accumulate *structural gaps* as product momentum outpaces organisational formalisation.

SECTION I: COMPANY STORYLINE & HISTORY

1. Origins: A Problem the Market Learned to Ignore (2019–2022)

Long before Aurex had a name, the problem it set out to solve was already visible in plain sight. Across India's fast-expanding cities, sewer and water networks were ageing faster than they could be mapped, inspected, or repaired. The result was predictable: overflows during monsoon peaks, chronic infiltration and inflow, silent leakages that never surfaced as incidents, and maintenance models built on complaint-response rather than continuous diagnostics. In most municipal ecosystems, the inspection function remained an afterthought - performed intermittently, outsourced episodically, and rarely converted into structured data.

Rohan Mehta and Ananya Kulkarni came to that conclusion from different directions. Rohan, trained as a robotics engineer, spent the better part of a decade working on industrial inspection automation in Europe, where pipeline diagnostics were treated as an engineering discipline rather than a reactive service. Ananya's exposure was deeper in embedded systems and applied AI: the gap between raw sensor signals and usable decisions. By 2021, both had begun to recognise the

same pattern in the global inspection industry: hardware had matured, but the field workflow remained slow, fragmented, and overly dependent on highly trained operators.

In late 2022, a short visit to Mumbai during monsoon season became the turning point. A sewer overflow near a busy arterial road, cleared by manual intervention and temporary pumping, was treated as routine by everyone involved. What unsettled them was not the incident itself but the absence of feedback loops. No consistent defect register, no digital trail, no traceable inspection history - and therefore no learning. A city was operating blind, and blindness had become normal.

2. Incorporation and First Principles (2023)

Aurex Dynamics Private Limited was incorporated in early 2023 in Mumbai with a simple first principle: inspection must generate structured intelligence, not just footage. The initial thesis combined three components- field robotics for mobility, acoustic sensing for leak and flow anomalies, and an analytics layer to standardise observations into repeatable defect categories that non-specialists could act upon.

The earliest months were spent in an unglamorous loop of prototypes and constraints. The company did not begin with a large R&D lab. It began with a modest co-working room in Andheri, a whiteboard full of mechanical sketches, and a founder's belief that iteration speed could compensate for limited capital. Aurex's first hires were not a large payroll- rather a small nucleus of engineers, supplemented by short-term specialists engaged for firmware and industrial design. In board language, this was described as a lean build model; in practice, it was the only way to move fast on budget.

During this period, Aurex also began forming informal relationships with municipal operators, contractors, and field supervisors. These early conversations shaped product design more than any lab test. The founders learned that the operating environment was brutal: variable pipe diameters, unpredictable debris load, poor access conditions, and maintenance teams under pressure to restore flow quickly rather than document root causes. A successful system would need to be rugged, forgiving, and quick to deploy—an engineering product with the temperament of a utility tool.

3. The Prototype Season: "If It Survives the Field, It's Real" (2023)

By mid-2023, Aurex produced its first working prototype crawler—an articulated chassis capable of navigating uneven pipe floors while carrying a camera payload and an acoustic sensor module. Early iterations used a combination of off-the-shelf components and custom firmware to translate acoustic signatures into signal features. The aim was not to perfect classification immediately; the aim was to build a repeatable data pipeline that could improve with exposure.

Two early pilot deployments were executed in Maharashtra under proof-of-concept arrangements. These pilots were valuable precisely because they were not controlled demonstrations. The crawler experienced real-world obstacles: silt banks, trapped plastics, intermittent flow surges, and unexpected junctions. Each failure produced learning, and each learning fed back into design modifications—chassis geometry tweaks, sealing improvements, sensor mounting redesigns, and firmware stability upgrades.

The first revenue receipts arrived quietly. They were not large contracts; they were service invoices for pilot support, site attendance, and basic reporting. In the narrative Aurex would later tell, these invoices were symbolic: proof that municipalities would pay for measurable inspection output. In operational terms, they also created a new pressure—customer expectations began to arrive before internal processes had fully matured.

By late 2023, Aurex's flagship platform was branded Aurex-Spectra™. The name was chosen to communicate a spectrum-based sensing approach, but it also served another purpose: it made the prototype feel like a product. The company began appearing at urban infrastructure exhibitions and industry meetups, presenting Aurex-Spectra as a proprietary alternative to imported inspection crawlers. In brochures and pitch decks, the phrase patent pending began to appear as shorthand for defensibility—a marketing device meant to signal seriousness in a crowded market.

4. Scaling Without Breaking: From Prototype to Deliverable (2024)

The first half of 2024 was defined by scale decisions. Demand was forming in pockets, but the founders understood the danger of scaling a prototype too early: a single field failure could collapse trust in the category, especially in municipal deployments where reputation travels faster than formal feedback. At the same time, Aurex was receiving more inbound inquiries than its small internal team could handle.

To expand production capacity, Aurex onboarded an OEM partner in Gujarat for chassis fabrication and assembly. The intent was to convert design iterations into repeatable manufacturing processes while retaining critical engineering control in-house. Simultaneously, the firmware and analytics layer continued to evolve through a mix of in-house work and specialised contract development.

This period saw Aurex build what many young engineering companies quietly rely upon: a web of fast-moving coordination. Field notes, design revisions, firmware builds, and issue logs moved quickly between founders, engineers, vendors, and pilot sites. The system worked because it was human—quick calls, rapid sharing of files, short feedback loops. The cost of this speed was formality: documentation discipline trailed behind product momentum.

In April 2024, Aurex initiated provisional patent filings covering two core concepts: a mechanical articulation mechanism intended for constrained pipe navigation, and an acoustic leak-detection method designed to classify anomalies in live networks. Internally, these were framed as protective filings—early fences around the most distinctive engineering claims. Externally, the company continued to speak in the language of innovation leadership: proprietary technology, patent-track R&D, and differentiated sensing.

By mid-2024, the company began limited international conversations, especially with infrastructure contractors operating in the Middle East. Aurex's view was pragmatic: export interest validated product relevance and improved investor story-telling. Initial shipments were small, structured as early partner deployments rather than mass export. But with export comes an additional layer of governance—classification, documentation, and compliance processes that rarely exist in early-stage operating rhythm.

5. The Municipal Reality: Trust, Tendering, and Politics of Proof (2024)

Municipal ecosystems reward proof more than promise. Aurex learned this quickly. In meetings with city engineers and field leadership, the questions were rarely about AI in the abstract. They were about downtime, serviceability, response times, spare parts availability, and the credibility of reporting. In other words, Aurex was no longer selling a product; it was selling reliability.

To increase credibility, Aurex began producing structured defect classification reports. A defect was no longer just a frame of video; it became a category, severity, location marker, and recommended action. Over time, these reports started to create internal data: a defect library, a signal history, and a growing belief that the inspection function could be systematised.

As the platform matured, Aurex also began receiving invitations to participate in larger municipal scopes—often as a technical partner to contractors. These opportunities brought a new kind of complexity: tender documentation, eligibility proofs, and the slow logic of public procurement. Aurex's technical story was strong. Its corporate scaffolding—policies, committee structures, contract registers—was still catching up.

6. The Trigger Event: Institutional Capital and Audit Readiness (2025)

In early 2025, Aurex received a formal venture capital term sheet and was shortlisted as a technical partner for a multi-year sewer rehabilitation tender involving digital monitoring and inspection services. The convergence of these two events forced a shift in mindset. What had

worked as a founder-led, speed-first company now needed to stand up to institutional scrutiny: ownership clarity, board oversight, repeatable processes, and a defensible compliance posture.

The Board resolved to commission an internal audit covering organisational maturity, legal and IP readiness, vendor and customer contracting discipline, and the overall sustainability of the operating model. This narrative section is drafted to establish the historical context and structural baseline against which management and legal review will be performed.

7. Corporate Structure Snapshot

Aurex Dynamics Private Limited operates as a promoter-led enterprise with centralised decision-making. The founders oversee product strategy, technical governance, and commercial relationships. Functional teams include Engineering (hardware, firmware, analytics), Field Operations (deployments and service), and Commercial (sales and partnerships). Outsourced partners support manufacturing, specialised firmware modules, and certain analytics workflows.

Aurex maintains a documented product platform roadmap and has initiated steps toward formal governance: board meetings on a quarterly cadence, basic policy documentation for IT and data handling, and a developing contract register. The company's operating philosophy remains execution-centric: decisions are made quickly, field feedback is prioritised, and iteration is continuous. This philosophy has been a competitive strength—while also creating identifiable areas where formality, traceability, and segregation may require strengthening.

8. Notable Milestones (Chronology)

January 2023: Company incorporated; initial product concept and field problem framing completed.

June 2023: First articulated crawler prototype achieved mobility in live network conditions.

October 2023: Pilot deployments conducted; first service invoices issued for field support and reporting.

November 2023: Aurex-Spectra™ branding introduced in external showcases and industry exhibitions (market-facing collateral).

April 2024: Provisional patent filings initiated for articulation mechanism and acoustic methodology.

July 2024: Manufacturing scaled via OEM partner onboarding; field deployments expanded across multiple wards.

November 2024: Early international partner deployments initiated; limited shipments executed through logistics partner.

January 2025: VC term sheet received; tender shortlisting triggers audit readiness programme.

9. Narrative Handoff to Management & Legal Review

The story of Aurex is, in many ways, the story of speed meeting scrutiny. A company born from a neglected infrastructure problem moved quickly to build a working system, earned early field credibility, and began attracting the attention of institutions that demand more than prototypes and promise. The sections that follow—Management and Legal—evaluate whether the corporate scaffolding beneath Aurex's momentum is commensurate with the scale it is now entering.

SECTION II: CORPORATE STRUCTURE & ORGANISATIONAL FRAMEWORK

10. Corporate Identity and Operating Context

Aurex Dynamics Private Limited is positioned as an engineering-led organisation operating at the intersection of technology development, infrastructure deployment, and service execution. The company's operating model reflects the realities of working in technically intensive environments where product performance, field execution reliability, and stakeholder coordination are equally critical.

The organisation was incorporated with the intent of delivering technically differentiated solutions to enterprise and institutional clients. Its activities span product design, system integration, deployment operations, and ongoing engagement with customers across geographically dispersed sites. As a result, Aurex Dynamics has developed an internal structure that prioritises functional specialisation, direct executive oversight, and operational responsiveness.

The company remains in a growth and consolidation phase, with its organisational structure shaped by early-stage constraints, founder involvement, and a focus on execution velocity rather than long-term institutionalisation.

11. Structural Philosophy and Design Principles

The organisational structure of Aurex Dynamics has been guided by a set of implicit design principles rather than a formally documented organisational doctrine. These principles include:

- Emphasis on functional expertise over matrixed coordination
- Concentration of strategic authority at the executive level
- Minimal layering between decision-makers and execution teams

- Preference for direct communication over procedural escalation

This design philosophy has resulted in a compact structure with clear functional ownership but limited structural redundancy. The organisation prioritises speed of decision-making and technical alignment, sometimes at the expense of formal separation of duties or institutional buffers.

12. Executive Leadership Composition

The executive leadership layer consists of two founder-led roles:

Chief Executive Officer (CEO)

The CEO holds primary responsibility for organisational leadership, commercial alignment, external stakeholder management, and overall operational coherence. This role serves as the central authority for business commitments, delivery assurance, and cross-functional arbitration.

Chief Technology Officer (CTO)

The CTO is responsible for the technical direction of the organisation, including product architecture, engineering standards, system performance, and long-term technology roadmap. The CTO oversees all engineering-related activities and maintains technical governance across development cycles.

The executive layer functions as both strategic leadership and operational coordination body, with limited delegation of cross-functional authority below this level.

13. Ownership and Capital Structure

Aurex Dynamics operates as a promoter-led private limited company. The ownership structure reflects early-stage capital formation and founder control:

- Founding promoters retain majority ownership and voting rights
- Angel investors hold minority, non-controlling equity positions
- An Employee Stock Option Plan (ESOP) pool exists to attract and retain early talent
- No institutional investors or board-level external governance structures are currently in place

This ownership configuration reinforces executive centrality in both strategic and operational matters.

14. Engineering Vertical

The Engineering Vertical represents the technical core of Aurex Dynamics. It is structured around domain-specific expertise and responsible for the full lifecycle of product development.

Scope of Responsibilities

- Mechanical and hardware system design
- Embedded firmware development
- Data processing and analytics logic
- System-level integration and validation
- Performance optimisation and testing

Engineering teams operate within defined technical domains, with coordination occurring through direct collaboration and executive review rather than formalised program management structures. Technical priorities are reviewed periodically with the CTO, aligning development efforts with commercial and deployment expectations.

15. Field Operations Vertical

The Field Operations Vertical manages the execution layer of the organisation, translating product capability into on-ground delivery.

Scope of Responsibilities

- Deployment planning and scheduling
- On-site execution and system operation
- Coordination with local partners and client representatives
- Field-level data capture and operational feedback

Field Operations is structured to respond dynamically to deployment requirements, with teams mobilised based on project timelines and site conditions. The vertical maintains a strong execution focus, with reporting routed directly to executive leadership.

16. Commercial Vertical

The Commercial Vertical is responsible for external market engagement and revenue generation activities.

Scope of Responsibilities

- Customer engagement and relationship management
- Identification and pursuit of opportunities
- Tender participation and proposal development
- Commercial negotiation and alignment with delivery capacity

Commercial activities are conducted with close executive involvement to ensure consistency between commitments and operational capability.

17. Central Administration Function

The Central Administration Function operates as a shared internal support unit across the organisation.

Scope of Responsibilities

- Internal coordination support
- Documentation and record maintenance
- Scheduling and internal communications
- Administrative facilitation across functions

The function is intentionally broad in scope, designed to provide flexible support rather than enforce formal process discipline.

18. Reporting Architecture

The reporting structure at Aurex Dynamics is strictly functional:

- Engineering reports to the CTO
- Field Operations reports to the CEO
- Commercial reports to the CEO
- Central Administration reports to the CEO

This architecture minimises reporting complexity but concentrates informational flow at the executive level.

19. Decision-Making Framework

Decision authority is functionally initiated but centrally reviewed:

- Technical decisions originate within Engineering
- Deployment decisions originate within Field Operations
- Commercial engagements originate within the Commercial Vertical

Decisions with cross-functional implications are escalated to the executive layer for review and approval.

20. Internal Coordination Model

Cross-functional coordination occurs through:

- Scheduled executive reviews

- Direct inter-functional communication
- Informal alignment discussions

The organisation does not maintain dedicated program management, integration offices, or cross-functional governance committees.

21. Knowledge and Continuity Context

Operational knowledge, stakeholder relationships, and execution context are primarily embedded within functional teams and the executive layer. Knowledge transfer mechanisms remain informal, relying on continuity of personnel rather than structured documentation systems.

22. Concluding Structural Perspective

Aurex Dynamics Private Limited reflects a structure optimised for early-stage execution, technical depth, and leadership proximity. While the organisational framework supports responsiveness and functional clarity, it also embeds a number of latent structural risks that may surface as scale, complexity, and stakeholder expectations increase.

The structure, as presently constituted, is coherent within its current operating envelope but would require deliberate redesign to support sustained growth, institutional resilience, and distributed leadership accountability.

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